Topic: Equal real Eigenvalues with multiplicity two

Question: Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 5 & -4 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 5 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} e^{5t} + c_3 \left(\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} te^{5t} + \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} e^{5t} \right)$$

B $$\vec{x} = c_1 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} e^{5t} + c_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} te^{5t}$$

C $$\vec{x} = c_1 \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} e^{5t} + c_3 \left(\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} te^{5t} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} e^{5t} \right)$$

D $$\vec{x} = c_1 \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} e^{5t} + c_3 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} te^{5t}$$

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Solution: C

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} 5 & -4 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 5 & -4 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 5-\lambda & -4 & 0 \\ 1 & -\lambda & 2 \\ 0 & 2 & 5-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (5-\lambda) \begin{vmatrix} -\lambda & 2 \\ 2 & 5-\lambda \end{vmatrix} - (-4) \begin{vmatrix} 1 & 2 \\ 0 & 5-\lambda \end{vmatrix} + 0 \begin{vmatrix} 1 & -\lambda \\ 0 & 2 \end{vmatrix}$$

$$|A - \lambda I| = (5-\lambda)[(-\lambda)(5-\lambda) - (2)(2)] + 4[(1)(5-\lambda) - (2)(0)]$$

$$|A - \lambda I| = (5-\lambda)(-5\lambda + \lambda^2 - 4) + 4(5-\lambda)$$

$$|A - \lambda I| = -25\lambda + 5\lambda^2 - 20 + 5\lambda^2 - \lambda^3 + 4\lambda + 20 - 4\lambda$$

$$|A - \lambda I| = -\lambda^3 + 10\lambda^2 - 25\lambda$$

Solve the characteristic equation to find the Eigenvalues.

$$-\lambda^3 + 10\lambda^2 - 25\lambda = 0$$

$$\lambda(\lambda^2 - 10\lambda + 25) = 0$$

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$$\lambda(\lambda - 5)(\lambda - 5) = 0$$

$$\lambda = 0,\ 5,\ 5$$

Let's deal with $\lambda_1 = 0$ first. We find

$$A - 0I = \begin{bmatrix} 5-0 & -4 & 0 \\ 1 & -0 & 2 \\ 0 & 2 & 5-0 \end{bmatrix}$$

$$A - 0I = \begin{bmatrix} 5 & -4 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 5 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -\frac{4}{5} & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{4}{5} & 0 \\ 0 & \frac{4}{5} & 2 \\ 0 & 2 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{4}{5} & 0 \\ 0 & 1 & \frac{5}{2} \\ 0 & 2 & 5 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -\frac{4}{5} & 0 \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

Turning these first two rows into equations gives

$$k_1 + 2k_3 = 0$$

$$k_2 + \frac{5}{2}k_3 = 0$$

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or

$$k_1 = -2k_3$$

$$k_2 = -\frac{5}{2}k_3$$

Let's choose $k_3 = -2$, which then gives $k_1 = 4$ and $k_2 = 5$.

$$\overrightarrow{k_1} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$$

Using this vector, one solution is

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix} e^{0t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$$

For the two remaining Eigenvalues, $\lambda_2 = \lambda_3 = 5$, we'll find

$$A - 5I = \begin{bmatrix} 5-5 & -4 & 0 \\ 1 & -5 & 2 \\ 0 & 2 & 5-5 \end{bmatrix}$$

$$A - 5I = \begin{bmatrix} 0 & -4 & 0 \\ 1 & -5 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$

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Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -5 & 2 \\ 0 & -4 & 0 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -5 & 2 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -5 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Turning these first two rows into equations gives

$$k_1 + 2k_3 = 0$$

$$k_2 = 0$$

or

$$k_1 = -2k_3$$

$$k_2 = 0$$

Let's choose $k_3 = -1$, which then gives $k_1 = 2$.

$$\overrightarrow{k_2} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

Other than the trivial solution $\overrightarrow{k_2} = (0,0,0)$, any other (k_1, k_2, k_3) set we find from $k_1 = -2k_3$ and $k_2 = 0$ will result in a vector that's linearly dependent with the $\overrightarrow{k_2} = (2,0,-1)$ vector we already found. So we can say that the Eigenvalue $\lambda_2 = \lambda_3 = 5$ produces only one Eigenvector.

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} e^{5t}$$

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Because we only find one Eigenvector for the two Eigenvalues $\lambda_2 = \lambda_3 = 5$, we have to use $\overrightarrow{k_2} = (2,0,-1)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_2}$$

$$\begin{bmatrix} 0 & -4 & 0 \\ 1 & -5 & 2 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$-4p_2 = 2$$

$$p_1 - 5p_2 + 2p_3 = 0$$

$$2p_2 = -1$$

The first and third equations both give $p_2 = -1/2$, so the system simplifies to just the second equation,

$$p_1 - 5\left(-\frac{1}{2}\right) + 2p_3 = 0$$

$$p_1 = -\frac{5}{2} - 2p_3$$

There are an infinite number of value pairs we can choose for $\overrightarrow{p_1} = (p_1, -1/2, p_3)$. Let's pick simple values, like $\overrightarrow{p_1} = (-1/2, -1/2, -1)$. But ideally, we'd like to simplify the solution by eliminating fractions and not leading with a negative value, so we'll multiply through $\overrightarrow{p_1}$ by -2 to rewrite it as

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$$\overrightarrow{p_1} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Then our second solution will be

$$\overrightarrow{x_3} = \overrightarrow{k_1} t e^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} te^{5t} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} e^{5t}$$

and the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} e^{5t} + c_3 \left(\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} te^{5t} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} e^{5t} \right)$$

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Question: Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ -1 & -2 & -2 \end{bmatrix} \vec{x}$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{t} + c_3 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} e^{t}$

B $\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} te^{-t}$

C $\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} e^{-t}$

D $\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} te^{-t} + c_3 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} e^{-t}$

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Solution: C

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ -1 & -2 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ -1 & -2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 2 & 1 \\ 1 & 1-\lambda & 1 \\ -1 & -2 & -2-\lambda \end{vmatrix}$$

$$|A - \lambda I| = -\lambda \begin{vmatrix} 1-\lambda & 1 \\ -2 & -2-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 1 \\ -1 & -2-\lambda \end{vmatrix} + 1 \begin{vmatrix} 1 & 1-\lambda \\ -1 & -2 \end{vmatrix}$$

$$|A - \lambda I| = -\lambda[(1-\lambda)(-2-\lambda) - (1)(-2)] - 2[(1)(-2-\lambda) - (1)(-1)]$$

$$+1[(1)(-2) - (1-\lambda)(-1)]$$

$$|A - \lambda I| = -\lambda(\lambda + \lambda^2) - 2(-1-\lambda) + (-1-\lambda)$$

$$|A - \lambda I| = -\lambda^2 - \lambda^3 + 2 + 2\lambda - 1 - \lambda$$

$$|A - \lambda I| = -\lambda^3 - \lambda^2 + \lambda + 1$$

Solve the characteristic equation to find the Eigenvalues.

$$-\lambda^3 - \lambda^2 + \lambda + 1 = 0$$

$$-(\lambda - 1)(\lambda + 1)^2 = 0$$

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$$\lambda = -1,\ -1,\ 1$$

Let's deal with $\lambda_1 = 1$ first. We find

$$A - (1)I = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 1-1 & 1 \\ -1 & -2 & -2-1 \end{bmatrix}$$

$$A - (1)I = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -1 & 2 & 1 \\ 1 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 1 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 2 & 2 \\ 0 & -4 & -4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & -4 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Turning these first two rows into equations gives

$$k_1 + k_3 = 0$$

$$k_2 + k_3 = 0$$

or

$$k_1 = -k_3$$

$$k_2 = -k_3$$

Let's choose $k_3 = -1$, which then gives $k_1 = 1$ and $k_2 = 1$.

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$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Using this vector, one solution is

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{t}$$

For the two remaining Eigenvalues, $\lambda_2 = \lambda_3 = -1$, we'll find

$$A - (-1)I = \begin{bmatrix} -(-1) & 2 & 1 \\ 1 & 1-(-1) & 1 \\ -1 & -2 & -2-(-1) \end{bmatrix}$$

$$A - (-1)I = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ -1 & -2 & -1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ -1 & -2 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Turning this first row into an equation gives

$$k_1 + 2k_2 + k_3 = 0$$

or

$$k_1 = -2k_2 - k_3$$

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Let's choose $k_2 = 0$ and $k_3 = -1$, which then gives $k_1 = 1$.

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

We could also choose $k_2 = -1$ and $k_3 = 0$, which then gives $k_1 = 2$.

$$\overrightarrow{k_3} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

So we can say that the Eigenvalue $\lambda_2 = \lambda_3 = -1$ produces two linearly independent Eigenvectors.

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t} \qquad\qquad \overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-t} \qquad\qquad \overrightarrow{x_3} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} e^{-t}$$

Therefore, the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} e^{-t}$$

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Topic: Equal real Eigenvalues with multiplicity two

Question: Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & 3 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{3t}$$

B $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^{3t}$$

C $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{3t} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{3t} \right)$$

D $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{3t} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^{3t} \right)$$

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Solution: C

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} 3 & 1 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 3 & 1 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 3 - \lambda & 1 & 0 \\ 0 & 4 - \lambda & -1 \\ 0 & 0 & 3 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (3 - \lambda) \begin{vmatrix} 4 - \lambda & -1 \\ 0 & 3 - \lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & -1 \\ 0 & 3 - \lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & 4 - \lambda \\ 0 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (3 - \lambda)[(4 - \lambda)(3 - \lambda) - (0)(-1)]$$

$$|A - \lambda I| = (3 - \lambda)(4 - \lambda)(3 - \lambda)$$

Solve the characteristic equation to find the Eigenvalues.

$$(3 - \lambda)(4 - \lambda)(3 - \lambda) = 0$$

$$\lambda = 4,\ 3,\ 3$$

Let's deal with $\lambda_1 = 4$ first. We find

$$A - 4I = \begin{bmatrix} 3 - 4 & 1 & 0 \\ 0 & 4 - 4 & -1 \\ 0 & 0 & 3 - 4 \end{bmatrix}$$

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$$A - 4I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Turning these first two rows into equations gives

$$k_1 - k_2 = 0$$

$$k_3 = 0$$

or

$$k_1 = k_2$$

$$k_3 = 0$$

Let's choose $k_2 = 1$, which then gives $k_1 = 1$ and $k_3 = 0$.

$$\vec{k_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Using this vector, one solution is

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t}$$

$$\vec{x_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{4t}$$

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For the two remaining Eigenvalues, $\lambda_2 = \lambda_3 = 3$, we'll find

$$A - 3I = \begin{bmatrix} 3-3 & 1 & 0 \\ 0 & 4-3 & -1 \\ 0 & 0 & 3-3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Turning these first two rows into equations gives

$$k_2 = 0$$

$$k_3 = 0$$

Let's choose $k_1 = 1$, which then gives

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

So we can say that the Eigenvalue $\lambda_2 = \lambda_3 = 3$ produces only one Eigenvector.

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t}$$

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Because we only find one Eigenvector for the two Eigenvalues $\lambda_2 = \lambda_3 = 3$, we have to use $\overrightarrow{k_2} = (1,0,0)$ to find a second solution.

$$(A - \lambda_2 I)\overrightarrow{p_1} = \overrightarrow{k_2}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$p_2 = 1$$

$$p_2 - p_3 = 0$$

or

$$p_2 = 1$$

$$p_3 = 1$$

Let's choose $p_1 = 0$, which then gives

$$\overrightarrow{p_1} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Then our second solution will be

$$\overrightarrow{x_3} = \overrightarrow{k_2} t e^{\lambda_2 t} + \overrightarrow{p_1} e^{\lambda_3 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{3t} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{3t}$$

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and the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{3t} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{3t} \right)$$

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